MINA: Do frozen DNA encoders beat sequence composition? A homology-aware linear-probe benchmark

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Abstract

Frozen DNA encoders are widely used as feature extractors, but fine-tuning hides what their fixed embeddings already make linearly accessible. MINA is a lightweight benchmark that asks whether those embeddings beat sequence composition, with two linear probes on 3,244 human genes: 5-way classification into coarse protein families, and regression onto each gene's GenePT text-summary embedding. Mostly, they do not. On coding sequence (CDS) under a homology-aware split, the validation-selected encoder, NT-v2, recovers coarse family signal at about the level of nucleotide composition (macro-F1 0.693 vs. 0.639; Holm $p = 0.108$) but does not reach amino-acid composition (0.767) or the protein language model ESM-2 (0.954). On GenePT, DNABERT-2 beats nucleotide composition (R^2 0.065 vs. 0.039) but not amino-acid composition (0.080). Evaluation choices move scores as much as the encoder does: pooling alone takes NT-v2 from 0.504, below nucleotide composition, to 0.698; a mostly noncoding transcription-start-site window, where any signal is likely regulatory rather than family, drops NT-v2 from 0.694 to 0.300 on the same genes; and a random split inflates it from 0.693 to 0.801 through homologous genes leaking into the test set. Frozen DNA encoders thus recover little from coding sequence beyond nucleotide composition, and fair comparisons must use composition baselines and fix pooling, substrate, and split.

Availability: code, splits, and artifacts: github.com/Austin-Senna/dna_to_text; sequences, features and predictions: Zenodo, doi:10.5281/zenodo.23031890.

1 Introduction

Pretrained DNA encoders such as DNABERT-2 (Zhou *et al.*, 2024), Nucleotide Transformer (Dalla-Torre *et al.*, 2025), GENA-LM (Fishman *et al.*, 2025), and HyenaDNA (Nguyen *et al.*, 2023) are increasingly used as genomic feature extractors, yet their embeddings, despite differing tokenisation, architecture, and pretraining corpus, are often treated as interchangeable summaries of biological sequence. Strong downstream performance after fine-tuning does not reveal what the encoder already represents; linear evaluation (Zhang *et al.*, 2016; Alain and Bengio, 2016), training only a linear readout (a *probe*) on frozen features, asks that narrower question.

We ask two questions of frozen DNA embeddings: do they carry protein-family signal beyond what sequence composition already provides, and do they carry gene function as described in text? To answer them we introduce MINA (Model Interrogation of Nucleotide Architectures), a lightweight linear-evaluation benchmark for frozen DNA encoders on 3,244 human protein-coding genes (Figure 1). One probe classifies each gene into one of five broad protein families; the other regresses its embedding onto the gene's GenePT vector, a 1,536-d OpenAI text embedding of the gene's short NCBI summary of what it does (Chen and Zou, 2024). Both read from two substrates: the coding sequence (CDS), which carries protein-domain signal directly, and a long transcription-start-site (TSS) window, which is mostly noncoding.

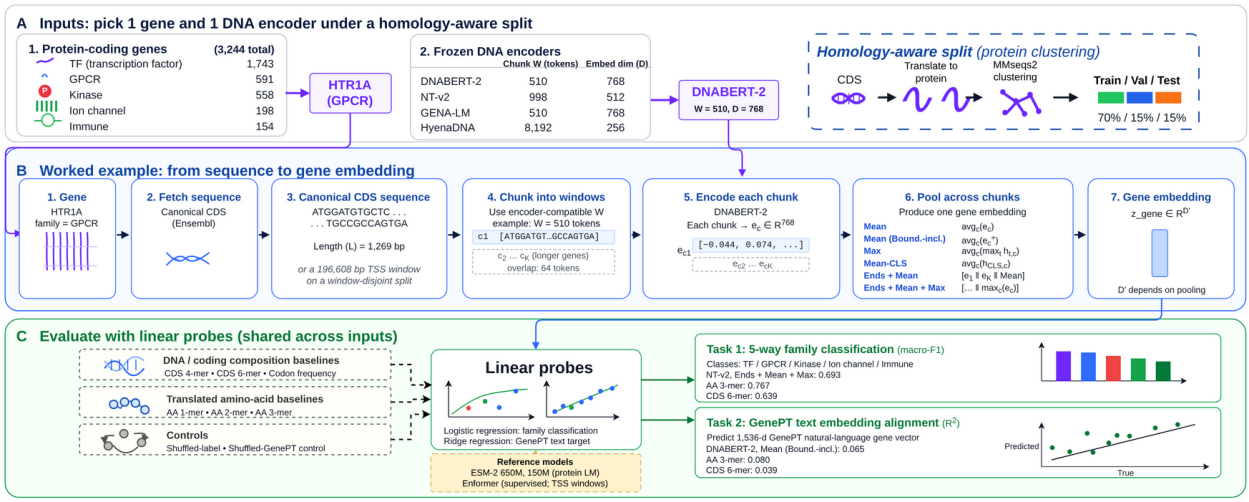


Figure 1: Study design. (A) Genes, encoders (W: content tokens per chunk), and the homology-aware split. (B) A gene's CDS is chunked, encoded, and pooled. (C) Linear probes against composition baselines, controls, ESM-2, and Enformer; printed scores are CDS test values for the encoder, pooling, and k chosen on validation.

Answering these questions fairly requires three controls. Frozen DNA-encoder embeddings probed on regulatory tasks offer little advantage over conventional models trained on one-hot sequence (Tang *et al.*, 2025), so every probe is anchored by nucleotide and translated amino-acid composition, with ESM-2 (Lin *et al.*, 2023) as a protein-level upper bound and a supervised Enformer comparator (Avsec *et al.*, 2021) on TSS windows. Homology between train and test inflates genome-trained models (Rafi *et al.*, 2025), echoing homology-aware splitting in protein machine learning (Li and Godzik, 2006; Rost, 1999; Teufel *et al.*, 2023), so genes are split by protein similarity with no close homologs shared between training and test. The pooling rule that turns chunk embeddings into one gene vector, usually left at a default, is varied across six rules.

Existing DNA-LM benchmarks do not combine these controls. GUANinE (Robson and Ioannidis, 2024) and BEND (Marin *et al.*, 2024) evaluate DNA models on regulatory and genome-annotation tasks. Neither asks what DNA embeddings of coding sequence carry about the encoded protein; BEND restricts variant effects to noncoding regions because coding effects can be predicted from the protein. DART-Eval (Patel *et al.*, 2024) finds that on regulatory DNA, DNA language models do not clearly beat simpler baselines; MINA extends that test to coding sequence, with composition and protein-level anchors. GUE (Zhou *et al.*, 2024) ranks encoders after fine-tuning, with no non-LM baselines. None anchors DNA encoders against amino-acid composition and a protein language model, the comparison MINA is built around.

Our contributions are threefold:

(i) MINA, a linear-evaluation benchmark for frozen DNA encoders, anchored by composition and a protein language model under a homology-aware split;

(ii) evidence that, under linear probes, frozen DNA encoders recover coarse protein-family signal from coding sequence at about the level of nucleotide composition, do not reach amino-acid composition, and recover little text-described function;

(iii) a quantification of how pooling, substrate, and train/test split each change a benchmark's conclusion.

2 Methods

MINA fixes a task and varies how it is evaluated. The task (§2.1) is a pair of linear probes on frozen DNA embeddings of 3,244 human genes, read against composition and protein-level anchors,

as shown in Figure 1. The evaluation axes (§2.2) are the pooling rule (Table 1), the input substrate, and the train/test split.

2.1 The benchmark

Genes and targets We curated 3,244 human protein-coding genes from five functional families: transcription factors (1,743), G-protein-coupled receptors (591), protein kinases (558), ion channels (198), and immune receptors (154). Genes were selected by matching each family's name against the HUGO Gene Nomenclature Committee (HGNC) gene-group annotations (Seal *et al.*, 2023), intersected with the GenePT v2 gene-symbol set (Chen and Zou, 2024); genes matching more than one family were assigned first-family-wins in the order kinase → transcription factor → ion channel → GPCR → immune receptor. Canonical coding sequences (CDS) were retrieved from Ensembl (Harrison *et al.*, 2024). Each gene carries two targets: its family label and its 1,536-d GenePT vector.

Encoders We compared four frozen self-supervised DNA encoders on canonical CDS. Three are BERT-style masked-language models with k-mer or byte-pair tokenisation: **DNABERT-2** (Zhou *et al.*, 2024), **Nucleotide Transformer v2 multi-species 100M** (Dalla-Torre *et al.*, 2025) (hereafter NT-v2) and **GENA-LM base** (Fishman *et al.*, 2025). The fourth, **HyenaDNA large** (Nguyen *et al.*, 2023), is an attention-free causal single-nucleotide model. The four span 6.6 M–117 M parameters and 512–2,048-token (up to 1M-nucleotide, for HyenaDNA) context windows; full checkpoints, tokenisation, and architecture details are given in Appendix A.2. All encoder weights were frozen.

Probes Each probe asks what a linear readout recovers from a frozen embedding. For family classification we trained multinomial logistic regression, scored by macro-F1, which weights each family equally, and by Cohen's κ (Vieira *et al.*, 2010), which weights each gene equally after discounting chance agreement. For GenePT we trained ridge regression to predict all 1,536 dimensions of each gene's GenePT vector, and report R^2 averaged across dimensions, where averaging weighs every dimension equally so a few high-variance dimensions cannot inflate the score. Cosine similarity between predicted and true vectors is a secondary measure. Every choice is made on the validation genes: the regularisation strength for every probe, the composition k, and, for the DNA encoders, the pooling rule and the encoder. We then refit on training and validation genes together and score once on the held-out test genes (grids in Appendix D.1).

Anchors Our probes have four reference points: **Composition**: parameter-free k-mer counts (Appendix Table A2), which test whether an encoder recovers more than sequence composition. Nucleotide k-mer counts are the floor: like the encoders, they ignore reading frame. On CDS, k is selected on validation ($k = 6$ for family classification). In-frame codon counts add the reading frame; amino-acid k-mer counts on the translated protein set the higher bar of protein composition. **Chance**: shuffling family labels or GenePT targets in train+val only should drive test scores to chance, which checks the pipeline for target leakage. **Protein upper bound**: ESM-2 (Lin *et al.*, 2023) (650 M) on the protein sequence, mean-pooled over residues. **Supervised regulatory comparator**: Enformer (Avsec *et al.*, 2021) on TSS windows (§2.2). It is trained on regulatory tracks, not family labels, so it is a reference point rather than a bound.

2.2 Evaluation axes

Pooling Encoders read long genes in chunks that fit their context windows, so each gene must be reduced to one vector, and that reduction can change what a probe sees. Parameter-free rules then reduce the chunks (Table 1): standard reductions for contextual embeddings (Devlin *et al.*, 2019; Reimers and Gurevych, 2019), two that also keep the first and last chunks, and, for TSS windows,

TSS-Anchored, which keeps only the chunk nearest the TSS.

Table 1: Pooling rules. Each reduces an encoder's chunk token embeddings to one gene vector; ‖ is concatenation. CLS is the summary token BERT-style tokenizers prepend to each input (Devlin *et al.*, 2019); HyenaDNA has none.

Rule	**Definition**
Mean	mean over tokens in each chunk, then over chunks
Mean (Boundary-including)	Mean, including the CLS/SEP boundary tokens
Max	max over tokens in each chunk, then mean over chunks
Mean-CLS	CLS token of each chunk, mean over chunks
Ends + Mean	first chunk ‖ last chunk ‖ Mean
Ends + Mean + Max	Ends + Mean ‖ max over chunks
TSS-Anchored	the chunk nearest the TSS (TSS windows only)

Substrate Two substrates carry the same gene. **CDS**: the coding sequence encodes the protein, so it carries the family label directly. **TSS window**: 196,608 bp of Ensembl sequence centred on each gene's strand-specific transcription start site, mostly noncoding, asking whether the label can also be read from the sequence around the gene's start. On TSS windows we ran all four encoders, the pooling rules except Mean (Boundary-including), and a matched 4-mer baseline; Enformer is read both as its whole-window trunk mean and over its central ∼2,048 bp.

Split A **random split** assigns genes individually, stratified by family, and lets closely related genes (paralogs, copies of a gene created by duplication) fall on both sides of the partition: 56% of its test genes have a relative in training or validation, so a probe can score by recognising near-copies (Appendix C.4). We therefore evaluate on two homology-aware splits and keep the random split as a control (§3.5). The **homology split**, primary for CDS, translates each canonical CDS, clusters the proteins with MMseqs2 (Steinegger and Söding, 2017) at 40% identity and 80% bidirectional coverage, above Rost's "twilight zone" (Li and Godzik, 2006; Rost, 1999), and assigns whole clusters, largest first, to one partition in 70/15/15 proportions stratified by family (Appendix Table A1); no test gene then shares a cluster with a training gene. The **disjoint split**, primary for TSS, additionally groups genes whose windows overlap on the genome, since windows of genes in different partitions can otherwise share sequence. Because MMseqs2 compares genes to a cluster representative rather than pairwise, a few close pairs still straddle the split; test genes with such a partner in training or validation (or, on the disjoint split, an overlapping window) are dropped from scoring, leaving 459 CDS and 465 TSS test genes. The homology split is repeated with four seeds for assigning clusters to partitions (Appendix Table C1; settings in Appendix D.1).

3 Results and Discussion

Differences are paired bootstrap estimates over groups of related test genes, with 95% intervals (Appendix Table C3; per-cell intervals in Appendix Tables B1 and B2; split-seed ranges in Appendix Table C1). Four comparisons are primary and Holm-corrected together: the selected encoder against the nucleotide k-mer and against the amino-acid k-mer, ESM-2 against the selected encoder, all on the homology split (§3.1), and NT-v2 on CDS against the TSS window of the same genes on the disjoint split (§3.4). We fixed the four after seeing the submitted results on these test genes, so they are not pre-registered (Appendix E). The remaining sections quantify how much pooling, substrate and split move scores, with uncorrected p-values; the pooling ranges in §3.3 are point estimates, not tests.

3.1 CDS encoders reach nucleotide composition, not amino-acid composition

If frozen DNA encoders capture protein-level signal, they should beat nucleotide composition on CDS; if that signal goes no further than composition, they should not beat amino-acid composition.

The selected encoder does not significantly beat nucleotide composition, and no encoder beats amino-acid composition. The validation-selected encoder, NT-v2, beats 6-mer composition by +0.054 macro-F1 [−0.011, +0.126], not significant after Holm correction ($p = 0.108$), and trails amino-acid 3-mer composition (−0.075 [−0.137, −0.001]). ESM-2, reading the protein directly, sits +0.261 [+0.200, +0.322] above NT-v2. GENA-LM scores highest on test (0.710) but also does not reach amino-acid composition (0.767). DNABERT-2 and HyenaDNA sit at the nucleotide floor (−0.011 and −0.006; Figure 2; Appendix Table B1).

We read this as NT-v2 pointing above nucleotide composition but not reliably (Holm-adjusted $p = 0.108$) and below translation, GENA-LM scoring similarly on test though it was not the validation pick, and DNABERT-2 and HyenaDNA recovering nothing beyond composition. Translated amino-acid counts use the reading frame the encoders are never given, and ESM-2 marks how much more a protein-native model reads.

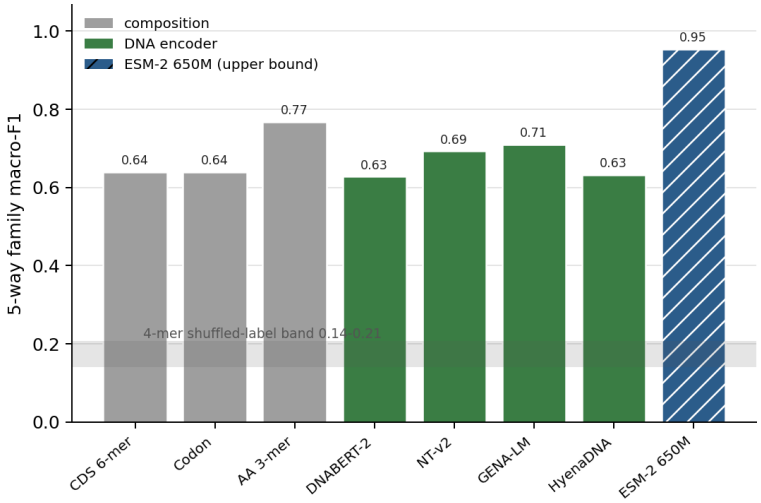


Figure 2: 5-way family classification macro-F1 on CDS: composition baselines (grey), DNA encoders at their validation-selected pooling (green), and ESM-2 650M on the protein sequence (blue, hatched). The nucleotide and amino-acid k are chosen on validation.

3.2 Encoders recover little text-described function, and none beyond amino-acid composition

If the family signal reflected gene function beyond protein composition, it should transfer to a text-derived function target. It does not. The best encoder, DNABERT-2, reaches GenePT R^2 0.065: above nucleotide 6-mer composition (+0.026 [+0.018, +0.034]), but at or below amino-acid 3-mer composition (−0.015 [−0.031, +0.000]), and ESM-2 sits +0.096 above amino-acid composition (Figure 3; Appendix Table B2).

Part of the weak alignment is the target itself. GenePT vectors inherit the geometry of English gene summaries, which sequence does not determine, so all vectors point in similar directions. Cosine similarity therefore stays high even for shuffled targets and cannot separate the features; R^2 can, which is why we report it, but even ESM-2 reaches only 0.176. Stronger alignment may require training it explicitly, as DNA–text models such as Omni-DNA do (Li *et al.*, 2025, 2026).

3.3 Pooling can move an encoder from below the composition floor to level with it

If pooling were a neutral default, an encoder's score would not depend on it. It does: NT-v2 scores macro-F1 0.504 with Max and 0.698 with Mean (Boundary-including), against 0.639 for the

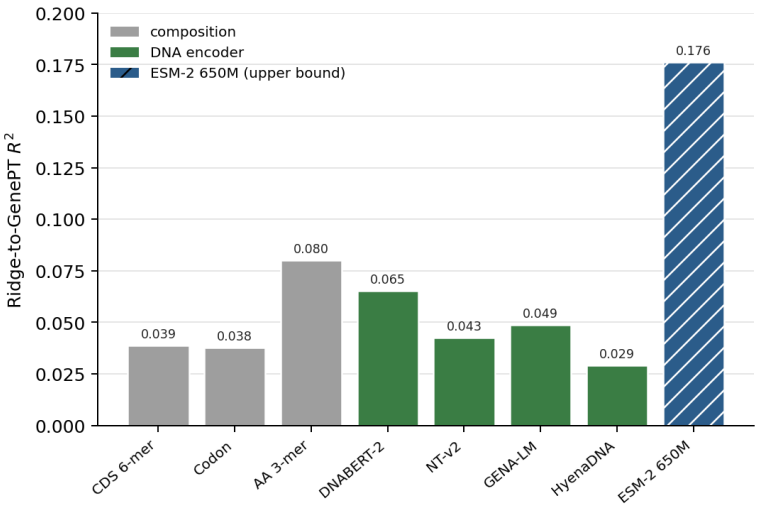


Figure 3: Ridge-to-GenePT R^2 on CDS, averaged over the 1,536 GenePT dimensions, for the same features and colours as Figure 2. The nucleotide and amino-acid k are chosen on validation for this probe.

validation-selected 6-mer, so pooling alone moves it from well below the composition floor to level with it. Pooling moves a single encoder by 0.159 to 0.196 macro-F1, well above the 0.041 to 0.087 that separates the four encoders at any one rule they all share, and Max is the weakest rule for every encoder. Because the best rule is encoder-specific, each encoder is reported at its validation-selected rule (Figure 4; full matrix in Appendix Table B5). Pooling is therefore a first-order design choice: frozen-encoder comparisons need explicit pooling controls, not a single default.

The rules that add the first and last chunks are mostly a regularisation effect: 81.9 to 99.0% of genes fit one window, where Ends + Mean is three copies of Mean. Against that control at matched regularisation, only GENA-LM keeps a lead (+0.086 [+0.017, +0.153]; Appendix Table C3).

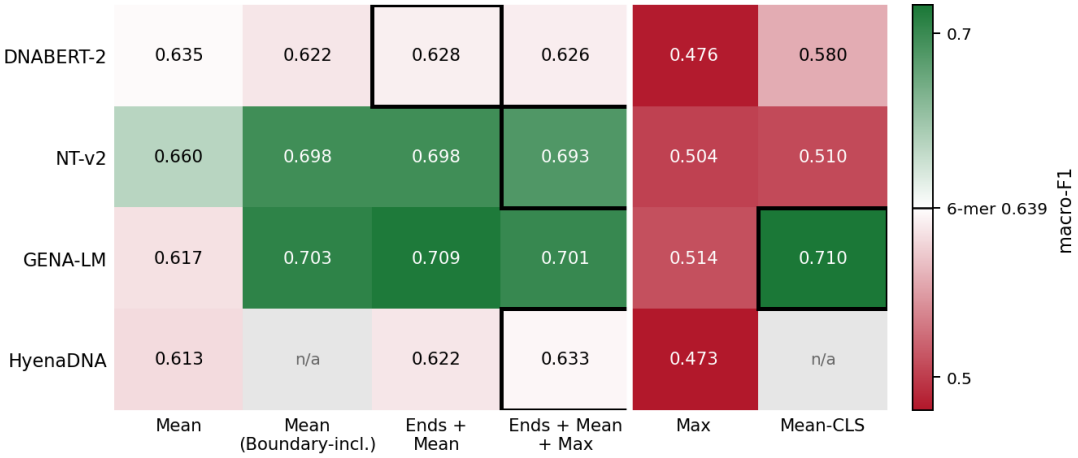


Figure 4: Macro-F1 across all encoder-pooling cells (CDS). Colour is centred on the validation-selected CDS 6-mer (white): green above, red below, darker further away. Boxes mark each encoder's validation-selected rule; n/a: the rule reads a trained boundary token, which HyenaDNA lacks.

3.4 The TSS window carries little of the family label

The TSS window is mostly promoter, intronic, and intergenic sequence (about 1% is the gene's own CDS; Appendix C.3), so any signal should be regulatory rather than family. The disjoint split, which removes the window overlap that 46.2% of homology-split test genes have with a training or

validation gene, lowers DNABERT-2 and NT-v2 but not the other models (Appendix Table C2). We read the window two ways, averaged over its whole length (§3.4.1) and at the TSS (§3.4.2), and Enformer the same two ways.

3.4.1 Whole-window pooling

If the window carried the family label, whole-window averages should approach CDS scores on the same genes; if it carries only promoter composition, they should sit near the TSS 4-mer. The second holds. The encoder selected on TSS validation, DNABERT-2, reaches macro-F1 0.258, below the TSS 4-mer (0.281) but above chance (shuffled-label median 0.167); the best encoder on test, GENA-LM, reaches 0.340. Enformer, trained on regulatory tracks and averaged over its output bins, reaches 0.413. Every encoder falls far below its CDS score on the same test genes (CDS minus TSS +0.267 to +0.393; Appendix Table C3). The window has no reading frame, so amino-acid composition is undefined and the 4-mer is the only composition anchor; GenePT R^2 is below zero for every TSS feature, so the window is reported on family macro-F1 only (Appendix Figure B1).

3.4.2 TSS-Anchored pooling

If a family signal sits near the TSS and whole-window averaging dilutes it, the TSS-Anchored readout (Methods) should recover it. It does not, reliably. All four encoders gain a little from reading at the TSS (+0.012 to +0.050 macro-F1), but every paired interval includes zero; the largest gain, NT-v2's, is +0.050 [−0.005, +0.106], and Enformer's central readout matches its whole-window mean (−0.001). The composition of the anchored chunk alone does about as well: for three of four encoders the anchored score is within interval of the chunk's own 4-mer or 6-mer baseline (diamonds in Figure 5; −0.049 to +0.015), and only GENA-LM beats it (+0.104 [+0.044, +0.161]). What the anchored readout recovers is therefore mostly the local composition around the TSS, not a diluted family signal, and every TSS result stays far below CDS (Appendix Tables B3 and C5).

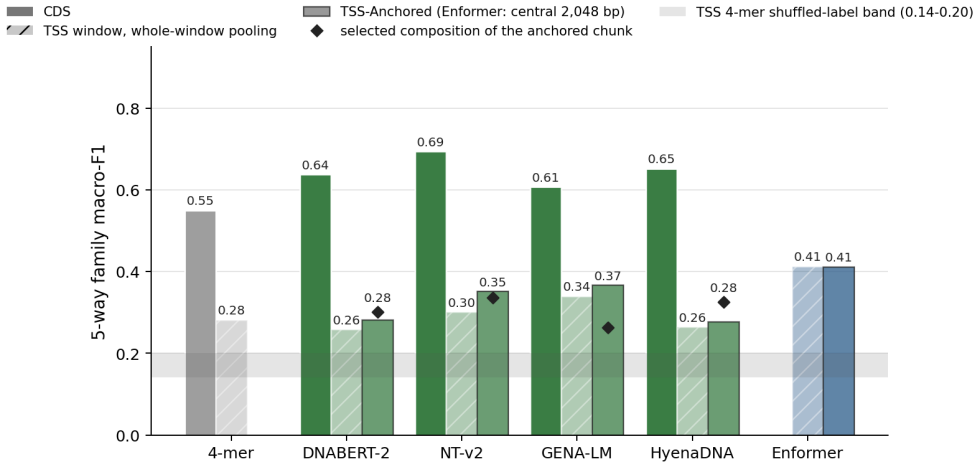


Figure 5: Family macro-F1 on CDS, the whole TSS window, and the TSS-Anchored chunk, all on the disjoint split. Enformer: whole-window mean and central 2,048 bp readout. Diamonds: the anchored chunk's validation-selected composition baseline. GenePT R^2 for the same cells is in Appendix Figure B1.

We interpret the TSS drop as a mismatch between substrate and target. The same frozen encoders recover the family label from CDS, so they can represent it; what the window lacks is its source, since only about 1% of it is the gene's own CDS. Consistent with this, reading at the TSS adds no reliable signal over the whole window, and no readout, including the supervised Enformer, approaches CDS. Whether a learned attention pooling over frozen token embeddings, or fine-tuning,

could find the coding exons inside the window, and whether a regulatory target can be read from it, we leave to future work.

3.5 Random splits inflate the TSS window more than CDS

If a random split leaks homologous genes into the test set, scores should be higher there than on each arm's primary split. Re-running every comparator on a random split measures the gap. The central reading survives: amino-acid composition leads the best DNA encoder on both splits (0.908 vs. 0.801 random, 0.767 vs. 0.693 homology), and ESM-2 leads throughout. What changes is the level, and unevenly. Every model and baseline scores higher on the random split except GENA-LM on CDS, which ties (Appendix Table B4). Counting only the margin above chance (macro-F1 0.167 under shuffled labels), NT-v2 keeps 83% of its random-split margin on CDS under the homology split but only 42% on the TSS window under the disjoint split, so a random split inflates the regulatory arm far more than the coding arm (Figure 6; Appendix Table B4).

We read this as evidence that the homology split, not a composition baseline alone, is what separates memorised similarity from transferable signal, the coding-sequence counterpart of the leakage Rafi *et al.* (2025) report for DNA-level expression models.

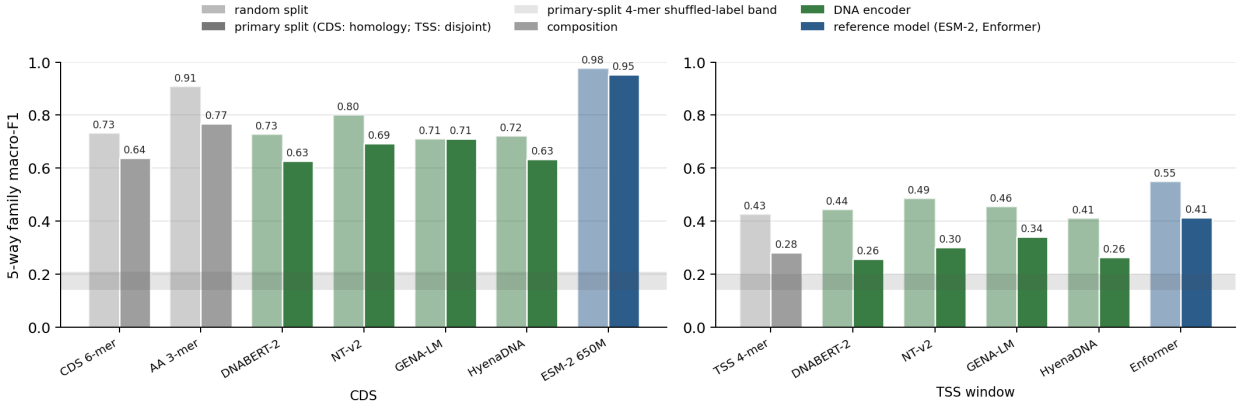


Figure 6: Family macro-F1 on the random-stratified split against each arm's primary split: CDS on the homology-aware split (left) and the TSS window on the disjoint split (right), every model at its validation-selected pool on each split. Shaded: each primary split's 4-mer shuffled-label band. GenePT R^2 per split is in Appendix Table B4.

4 Conclusion and Limitations

4.1 Conclusion

MINA fixes a task and varies how it is evaluated. On that task, read through linear probes, frozen DNA encoders recover coarse protein-family signal from coding sequence at about the level of nucleotide composition, do not reach translated amino-acid composition, and recover little of the text-described function in GenePT. How the task is evaluated changes the conclusions as much as which encoder is used: the pooling rule alone can move an encoder from well below the composition floor to level with it, a mostly noncoding substrate cuts the family signal far below CDS, and random splits inflate most scores through homologous genes.

We draw four recommendations for evaluating frozen DNA embeddings: compare every score against nucleotide and amino-acid composition; treat pooling as a reported, validation-selected choice rather than a default; match the substrate to the target; and split by homology, and by genomic interval when inputs are windows. The next steps are a learned pooling over frozen token embeddings, regulatory targets for the TSS window, and families beyond the five used here.

4.2 Limitations

Scope. MINA is small by design: cheap to repeat across encoders, pooling rules, substrates and splits, at the cost of power and breadth. It covers 3,244 human genes in five broad HGNC families, 99.9% of GenePT-covered genes in those families; the remaining ∼15,600 GenePT genes have no broad-family ontology, so a larger label set is a separate many-class study. The five families are a best case, large and domain-rich, and every comparison is within this one task with one chance floor, so the conclusions may not transfer to families with weaker signatures or to other class counts. The protocol is species-agnostic; GenePT and the HGNC families are not.

Labels and targets. Family labels come from broad HGNC group names, which add noise: 387 transcription-factor labels come only from zinc-finger groups outside the C2H2 class (a rule that also catches genuine factors such as GATA1–6), and 150 kinase labels are not protein kinases; masking either set leaves every primary test pointing the same way (Appendix Table C3). GenePT is the only text target and enters only the regression arm. It inherits literature bias, and 901 genes share a templated summary; masking them leaves the ordering unchanged. Macro-averaged R^2 depends on the GenePT basis; pooled R^2, which does not, keeps the encoder ordering, and retrieval@5 reorders the DNA encoders but keeps ESM-2 first (Appendix Table C4).

Readout. A linear probe on a frozen, pooled embedding is a lower bound on what the embedding contains: signal it finds is there, but signal it misses may still be recoverable by a nonlinear probe, learned pooling, or fine-tuning. The DNA encoders were pretrained on unlabelled genomes, ESM-2 on UniRef proteins, and Enformer on human and mouse regulatory tracks, so each may have seen our test sequences; none was trained on the family labels or GenePT targets.

Statistics. The test set is small: 459 CDS genes, only 22 of them immune receptors, so a paired macro-F1 interval spans about ±0.07 and smaller differences are not resolved, and macro-F1 lets those 22 genes count as much as the 245 transcription factors. The intervals cover only the sampling of test genes, not the uncertainty from hyperparameter, split-seed, or pooling choices, which we report separately (split seeds in Appendix Table C1, pooling in Appendix Table B5, the numerical backend in Appendix D.1). TSS-Anchored pooling was designed after seeing the whole-window results, so it is exploratory.

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A Dataset, Baselines and Encoders

Appendix A defines the dataset, baselines and encoders; Appendix B gives the full values behind every main-text figure and the complete pooling and Ridge grids; Appendix C reports confidence intervals and robustness checks; Appendix D covers code and settings; Appendix E lists what changed since submission. CDS tables use the homology-aware split; TSS tables and every CDS–TSS comparison use the disjoint split (both 70/15/15). Random-split tables say so. Every table uses the main text's validation-tuned probe protocol. A dagger (†) marks a digit that is not stable across numerical backends of the solver (Appendix D.1); no primary test is affected.

A.1 Dataset Composition and Baseline Definitions

Table A1: Curation. Each family's include patterns are matched, case-insensitively, against the HGNC `gene_group` field of protein-coding genes with an Ensembl ID; exclude patterns (in parentheses) remove a gene from that family. Match: genes matching; GenePT: also in GenePT; Kept: after first-family-wins assignment in the order shown (17 genes match two families); Train/Val/Test: the homology-aware split. Three olfactory receptors have no retrieved CDS, so GPCR enters the split with 591 and the dataset holds 3,244.

Family	Include (exclude) patterns	Match	GenePT	Kept	Train	Val	Test
Kinase	`\bkinase` (kinase inhibitor, kinase regulator, kinase substrate, pseudokinase)	564	558	558	391	83	84
TF	zinc finger, homeobox, basic helix-loop-helix, `\bbZIP\b`, forkhead box, high mobility group, nuclear receptor, T-box, SOX transcription, ETS transcription, transcription factor (transcription factor binding, cofactor)	1,757	1,745	1,743	1,220	262	261
Ion channel	`\bchannel` (channel regulat, channel auxiliary, channel interacting)	201	200	198	138	30	30
GPCR	G protein-coupled receptor, adrenoceptor, 5-hydroxytryptamine receptor, dopamine receptor, muscarinic, opioid receptor, chemokine receptor, olfactory receptor	617	601	594	414	88	89
Immune	toll like receptor, `interleukin .* receptor`, `\bFc receptor`, NOD-like receptor, NLR family, killer cell immunoglobulin, T cell receptor, B cell receptor, C-type lectin domain, immunoglobulin like receptor (binding)	164	160	154	108	23	23

Table A2: Parameter-free composition baselines, each an L_1-normalised frequency vector run through the same probe recipe. Controls append raw log-length (and GC) terms and are never candidates for the headline composition pick: the length controls feed the composition-with-length sensitivity test (Table C3), and the chunk baselines are the composition of the same anchored chunk an encoder reads (Table C5).

Baseline	Feature	Dim
CDS 4-mer	Overlapping 4-nucleotide windows, frame-agnostic	256
CDS 6-mer	Overlapping 6-nucleotide windows	4,096
Codon	In-frame codons (frame 0, non-overlapping)	64
AA 1-mer	Amino-acid composition of translated CDS	20
AA 2-mer	Overlapping dipeptide frequencies	400
AA 3-mer	Overlapping tripeptide frequencies	8,000
Controls		
· + length	A nucleotide or AA k-mer above, plus $\log(1 + \text{CDS length})$	+1
Chunk 4-mer + GC	Anchored chunk: 4-mer, GC, two log-lengths	259
Chunk 6-mer	Anchored chunk: 6-mer	4,096

A.2 Encoder Architectures

DNABERT-2 (Zhou *et al.*, 2024) is a 117 M-parameter BERT-style model (`zhihan1996/DNABERT-2-117M`) with byte-pair encoding, 768-d hidden states, and a 512-token context window. **NT-v2** (Dalla-Torre *et al.*, 2025) (`InstaDeepAI/nucleotide-transformer-v2-100m-multi-species`) is an ESM-style masked-language model with fixed non-overlapping 6-mer tokens, 512-d hidden states, and a 2,048-token context window. **GENA-LM base** (Fishman *et al.*, 2025) (`AIRI-Institute/gena-lm-bert-base-t2t`) is a 110 M-parameter BERT-12L masked-language model with 768-d hidden states, 12 attention heads, a 32,000-token BPE vocabulary, and a 512-token context window corresponding to roughly 4.5 kb after BPE compression; the `t2t` checkpoint was trained on the human T2T assembly with 1000 Genomes SNP augmentation. **HyenaDNA large** (Nguyen *et al.*, 2023) (`LongSafari/hyenadna-large-1m-seqlen-hf`) is an attention-free causal next-nucleotide model based on Hyena sequence operators, with single-nucleotide tokens, 8 layers, 256-d hidden states, approximately 6.6 M trainable parameters, and a context window up to 1M nucleotides; HyenaDNA was pretrained on the human reference genome.

B Full Values

B.1 Full Values for Main-Text Figures

Each table below gives the exact values plotted in the corresponding main-text figure.

Table B1: Full values for Figure 2 (5-way family classification), each with its 95% cluster-bootstrap CI (1,000 resamples); the shuffled-label row carries its null band. Macro-F1 and κ order neighbouring encoders differently but agree that no DNA encoder reaches amino-acid composition. Bold: the best value in each score column outside ESM-2 (the upper bound) and the controls.

Source	Pool	F1 [95% CI]	ΔF1	κ [95% CI]	Acc.
Shuffled labels (median of 200)	—	0.167 [0.141, 0.208]	-0.472	-0.015	0.475
CDS 6-mer	—	0.639 [0.572, 0.697]	+0.000	0.593 [0.521, 0.653]	0.741
Codon	—	0.640 [0.569, 0.704]	+0.001	0.617 [0.550, 0.682]	0.767
AA 1-mer	—	0.697 [0.637, 0.761]	+0.058	0.679 [0.616, 0.739]	0.800
AA 2-mer	—	0.760 [0.698, 0.816]	+0.122	0.729 [0.668, 0.783]	0.832
AA 3-mer	—	**0.767 [0.703, 0.821]**	+0.129	**0.757 [0.698, 0.808]**	**0.847**
DNABERT-2	Ends + Mean	0.628 [0.557, 0.690]	-0.011	0.598 [0.536, 0.666]	0.749
NT-v2	Ends + Mean + Max	0.693 [0.629, 0.754]	+0.054	0.653 [0.594, 0.710]	0.780
GENA-LM	Mean-CLS	0.710 [0.647, 0.765]	+0.071	0.651 [0.590, 0.710]	0.782
HyenaDNA	Ends + Mean + Max	0.633 [0.571, 0.686]	-0.006	0.573 [0.503, 0.631]	0.730
ESM-2 650M	—	0.954 [0.923, 0.977]	+0.315	0.939 [0.907, 0.966]	0.961

Table B2: Full values for Figure 3 (Ridge-to-GenePT regression), each with its 95% cluster-bootstrap CI. Bold: the best value in each score column outside ESM-2 (the upper bound) and the controls.

Source	Pool	R^2 [95% CI]	Δ	Cos.
CDS 6-mer	—	0.039 [0.014, 0.057]	+0.000	0.916
Codon	—	0.038 [0.018, 0.048]	-0.001	0.916
AA 1-mer	—	0.022 [0.004, 0.033]	-0.017	0.915
AA 2-mer	—	0.059 [0.040, 0.073]	+0.021	0.918
AA 3-mer	—	**0.080 [0.058, 0.094]**	+0.042	**0.920**
DNABERT-2	Mean (Boundary-including)	0.065 [0.038, 0.089]	+0.026	0.919
NT-v2	Mean (Boundary-including)	0.043 [0.018, 0.057]	+0.004	0.917
GENA-LM	Ends + Mean	0.049 [0.024, 0.067]	+0.010	0.917
HyenaDNA	Ends + Mean	0.029 [0.006, 0.044]	-0.010	0.915
ESM-2 650M	—	0.176 [0.152, 0.191]	+0.137	0.929

Table B3: Full values for the CDS and whole-window bars of Figure 5: per-encoder classification (macro-F1, with κ) and Ridge-to-GenePT (R^2), CDS versus TSS window under whole-window pooling. Every row is on the disjoint split, so CDS and TSS score the same test genes; Δ is against the 4-mer of the same substrate, the only composition baseline defined on the window (Table B1 uses the validation-selected k on the homology split). On CDS the stronger baselines are defined too (6-mer 0.627, amino-acid 3-mer 0.762), and every encoder's macro-F1 stays below amino-acid composition, as on the homology split. Bold: the best DNA encoder's F1 and ΔF1 in each block, and its CDS R^2 and ΔR^2; TSS R^2 is below zero for every encoder and is not bolded.

Source	F1	ΔF1	κ	R^2	ΔR^2
Coding sequence (CDS)					
4-mer	0.550	+0.000	0.509	0.014	+0.000
DNABERT-2	0.638	+0.088	0.607	**0.048**	**+0.034**
NT-v2	**0.694**	**+0.144**	0.644	0.025	+0.011
GENA-LM	0.607†	+0.057†	0.537†	0.027	+0.013
HyenaDNA	0.652	+0.102	0.587	0.015	+0.001
TSS-centred window (196,608 bp)					
4-mer	0.281	+0.000	0.123	-0.034	+0.000
DNABERT-2	0.258	-0.024	0.104	-0.011	+0.023
NT-v2	0.300	+0.019	0.190	-0.013	+0.021
GENA-LM	**0.340**	**+0.058**	0.192	-0.020	+0.014
HyenaDNA	0.264	-0.017	0.113	-0.037	-0.003
Enformer (whole window)	0.413	+0.131	0.268	-0.006	+0.028

Table B4: Full values for Figure 6: random-stratified versus each arm's primary split (CDS homology-aware, TSS disjoint), by sequence context, with leakage deltas.

Source	F1 (rand)	F1 (prim.)	ΔF1	R^2 (rand)	R^2 (prim.)	ΔR^2
Coding sequence (CDS)						
CDS 6-mer	0.733	0.639	-0.094	0.190	0.039	-0.151
Codon	0.693	0.640	-0.054	0.173	0.038	-0.135
AA 2-mer	0.854	0.760	-0.094	0.204	0.059	-0.144
AA 3-mer	0.908	0.767	-0.141	0.278	0.080	-0.198
NT-v2	0.801	0.693	-0.108	0.192	0.043	-0.149
DNABERT-2	0.728	0.628	-0.101	0.210	0.065	-0.145
GENA-LM	0.710	0.710	-0.000	0.191	0.049	-0.142
HyenaDNA	0.721	0.633	-0.089	0.181	0.029	-0.152
ESM-2 650M	0.977	0.954	-0.023	0.350	0.176	-0.174
TSS window (196,608 bp; primary = genomic-interval-disjoint split)						
TSS 4-mer	0.427	0.281	-0.145	0.086	-0.034	-0.120
DNABERT-2	0.445	0.258	-0.187	0.129	-0.011	-0.140
NT-v2	0.487	0.300	-0.187	0.124	-0.013	-0.137
GENA-LM	0.456	0.340	-0.116	0.113	-0.020	-0.133
HyenaDNA	0.410	0.264	-0.146	0.091	-0.037	-0.128
Enformer (whole window)	0.550	0.413	-0.137	0.134	-0.006	-0.140

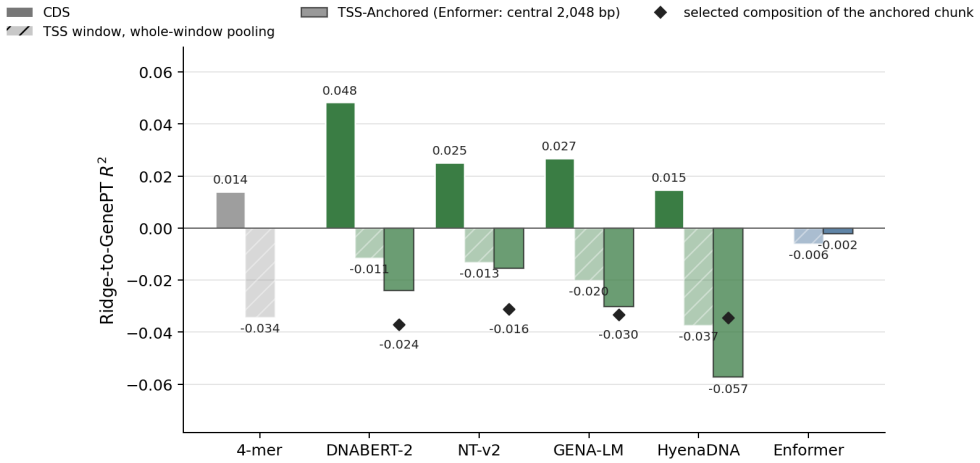


Figure B1: Ridge-to-GenePT R^2 for the cells of Figure 5: CDS, the whole TSS window, and the TSS-Anchored chunk, on the disjoint split. Every TSS feature scores below zero, so the probe predicts the text embedding worse than the training mean does; the main text therefore reports the TSS window on family macro-F1 only. CDS and whole-window values are in Table B3; TSS-Anchored values are printed on the bars.

B.2 Full pooling and Ridge matrices: primary vs. random split

Each readout is shown as one full-width matrix with the primary split (homology-aware for CDS, disjoint for TSS) beside its random-stratified mirror, so any encoder-pooling cell can be read across the two splits to gauge the homology-leakage inflation. Every cell's macro-F1 is higher on the random split except GENA-LM's CDS Mean-CLS, Ends + Mean and Ends + Mean + Max cells, and translated amino-acid composition matches or beats the best DNA encoder on both. The full classification matrix is Table B5 and the full Ridge matrix is Table B6, given on the following pages.

Table B5: Every cell for 5-way family classification: the primary split (left; CDS on the homology-aware split, TSS on the genomic-interval-disjoint split) versus the random-stratified split (right). Each split selects C on its own validation set. Mean and Mean (Boundary-including) differ only by the boundary tokens: for NT-v2 and GENA-LM they score alike wherever validation picks the same C, and on the primary CDS split validation picked a far larger C for Mean on a nearly flat curve, which costs it on test.

		Primary split			**Random-stratified split**		
Source	Pooling	Macro-F1	κ	Accuracy	Macro-F1	κ	Accuracy
Coding sequence (CDS; primary = homology-aware split)							
CDS 4-mer	—	0.6482	0.5905	0.7495	0.6852	0.7018	0.8090
CDS 6-mer	—	0.6386	0.5927	0.7407	0.7327	0.7314	0.8275
Codon	—	0.6397	0.6165	0.7669	0.6932	0.7154	0.8214
AA 1-mer	—	0.6970	0.6792	0.7996	0.7064	0.7214	0.8255
AA 2-mer	—	0.7603	0.7289	0.8322	0.8540	0.8261	0.8891
AA 3-mer	—	0.7674	0.7570	0.8475	0.9081	0.9040	0.9384
DNABERT-2	Mean	0.6351	0.6073	0.7538	0.7150	0.6976	0.8090
DNABERT-2	Mean (Boundary-including)	0.6217	0.5991	0.7495	0.7205	0.7043	0.8131
DNABERT-2	Max	0.4765	0.3899	0.6013	0.5875	0.5654	0.7228
DNABERT-2	Mean-CLS	0.5798†	0.5150†	0.6797†	0.6487	0.6693	0.7967
DNABERT-2	Ends + Mean	0.6276	0.5983	0.7495	0.7361	0.7250	0.8255
DNABERT-2	Ends + Mean + Max	0.6264	0.6005	0.7516	0.7284	0.7192	0.8234
NT-v2	Mean	0.6604†	0.6059†	0.7495†	0.8007	0.7947	0.8706
NT-v2	Mean (Boundary-including)	0.6983	0.6577	0.7843	0.8007	0.7947	0.8706
NT-v2	Max	0.5036	0.4987	0.7015	0.6038	0.6162	0.7659
NT-v2	Mean-CLS	0.5105	0.4806	0.6754	0.6041	0.5782	0.7331
NT-v2	Ends + Mean	0.6982	0.6633	0.7865	0.8108	0.8087	0.8789
NT-v2	Ends + Mean + Max	0.6928	0.6525	0.7800	0.8028	0.7871	0.8645
GENA-LM	Mean	0.6172†	0.5652†	0.7168†	0.7103	0.6954	0.8090
GENA-LM	Mean (Boundary-including)	0.7033	0.6450	0.7778	0.7103	0.6954	0.8090
GENA-LM	Max	0.5143	0.4120	0.6187	0.6002	0.5898	0.7474
GENA-LM	Mean-CLS	0.7100	0.6506	0.7821	0.6726	0.6864	0.8049
GENA-LM	Ends + Mean	0.7085	0.6597	0.7887	0.6627	0.6906	0.8111
GENA-LM	Ends + Mean + Max	0.7010	0.6584	0.7865	0.6984	0.7172	0.8234
HyenaDNA	Mean	0.6131	0.5595	0.7233	0.7025	0.6843	0.8049
HyenaDNA	Max	0.4730	0.3794	0.6122	0.5639	0.5304	0.7084
HyenaDNA	Ends + Mean	0.6218	0.5629	0.7233	0.7241	0.6894	0.8070
HyenaDNA	Ends + Mean + Max	0.6326	0.5727	0.7298	0.7214	0.6840	0.8029
ESM-2 650M	—	0.9536	0.9392	0.9608	0.9768	0.9745	0.9836
TSS-centred window (196,608 bp; primary = genomic-interval-disjoint split)							
TSS 4-mer	—	0.2814	0.1227	0.4817	0.4268	0.3503	0.6119
DNABERT-2	Mean	0.2986	0.1332	0.5075	0.4446	0.3634	0.6099
DNABERT-2	Max	0.2179	0.0962	0.5054	0.3940	0.2908	0.5688
DNABERT-2	Mean-CLS	0.2861	0.1131	0.4731	0.3963	0.2769	0.5195
DNABERT-2	Ends + Mean	0.2552	0.1452	0.4989	0.4223†	0.3225†	0.5914†
DNABERT-2	Ends + Mean + Max	0.2577	0.1043	0.4688	0.4137†	0.3305†	0.5749†
NT-v2	Mean	0.3004	0.1900	0.5419	0.4870	0.4217	0.6468
NT-v2	Max	0.2515	0.1062	0.4774	0.3842	0.3234	0.5934
NT-v2	Mean-CLS	0.2372	0.0763	0.4301	0.3107	0.2371	0.5791
NT-v2	Ends + Mean	0.2941	0.1486	0.4731	0.3904	0.2873	0.5647
NT-v2	Ends + Mean + Max	0.2873	0.1674	0.4839	0.3686	0.2789	0.5298
GENA-LM	Mean	0.3397	0.1916	0.5290	0.4557	0.3654	0.6160
GENA-LM	Max	0.2506	0.0690	0.4129	0.4168	0.3266	0.5975
GENA-LM	Mean-CLS	0.3398†	0.1523†	0.4323	0.4394†	0.3255†	0.5626†
GENA-LM	Ends + Mean	0.3085	0.1865	0.5204	0.4145	0.3114	0.5893
GENA-LM	Ends + Mean + Max	0.2479†	0.0990†	0.4301†	0.4387	0.3424	0.5975
HyenaDNA	Mean	0.2640	0.1132	0.4839	0.4150	0.3323	0.6119
HyenaDNA	Max	0.1947	0.0718	0.5097	0.3119	0.2310	0.5400
HyenaDNA	Ends + Mean	0.2638	0.1167	0.4645	0.3663	0.2407	0.5154
HyenaDNA	Ends + Mean + Max	0.2892	0.1730	0.5376	0.4103	0.2804	0.5606
Enformer (whole window)	—	0.4127	0.2677	0.5613	0.5498	0.4365	0.6427

Table B6: Every Ridge-to-GenePT cell: the primary split (left; CDS on the homology-aware split, TSS on the genomic-interval-disjoint split) versus the random-stratified split (right). Each split selects α on its own validation set.

		Primary split			**Random-stratified split**		
Source	Pooling	R^2 macro	Mean cosine	α	R^2 macro	Mean cosine	α
Coding sequence (CDS; primary = homology-aware split)							
CDS 4-mer	—	0.0320	0.9157	10000	0.1711	0.9304	1000
CDS 6-mer	—	0.0387	0.9165	10000	0.1900	0.9322	10000
Codon	—	0.0377	0.9165	1000	0.1732	0.9305	1000
AA 1-mer	—	0.0220	0.9147	10000	0.1662	0.9298	100
AA 2-mer	—	0.0595	0.9183	10000	0.2036	0.9333	10000
AA 3-mer	—	0.0802	0.9203	10000	0.2777	0.9399	10000
DNABERT-2	Mean	0.0651	0.9188	10000	0.2062	0.9336	10000
DNABERT-2	Mean (Boundary-including)	0.0652	0.9188	10000	0.2061	0.9336	10000
DNABERT-2	Max	0.0398	0.9163	10000	0.1700	0.9302	10000
DNABERT-2	Mean-CLS	0.0499	0.9174	10000	0.1880	0.9319	10000
DNABERT-2	Ends + Mean	0.0640	0.9188	10000	0.2101	0.9340	10000
DNABERT-2	Ends + Mean + Max	0.0599	0.9183	100000	0.2095	0.9339	10000
NT-v2	Mean	0.0426	0.9169	1000	0.1916	0.9322	1000
NT-v2	Mean (Boundary-including)	0.0426	0.9169	1000	0.1915	0.9322	1000
NT-v2	Max	0.0205	0.9147	10000	0.1382	0.9271	10000
NT-v2	Mean-CLS	0.0010	0.9126	10000	0.1168	0.9251	1000
NT-v2	Ends + Mean	0.0416	0.9168	10000	0.1918	0.9323	1000
NT-v2	Ends + Mean + Max	0.0426	0.9169	10000	0.1871	0.9318	10000
GENA-LM	Mean	0.0423	0.9167	10000	0.1824	0.9314	10000
GENA-LM	Mean (Boundary-including)	0.0423	0.9167	10000	0.1824	0.9314	10000
GENA-LM	Max	0.0223	0.9148	10000	0.1551	0.9288	10000
GENA-LM	Mean-CLS	0.0370	0.9162	10000	0.1885	0.9320	1000
GENA-LM	Ends + Mean	0.0488	0.9174	10000	0.1899	0.9321	10000
GENA-LM	Ends + Mean + Max	0.0493	0.9175	10000	0.1906	0.9322	10000
HyenaDNA	Mean	0.0263	0.9152	1000	0.1811	0.9312	100
HyenaDNA	Max	-0.0015	0.9124	10000	0.1294	0.9263	1000
HyenaDNA	Ends + Mean	0.0290	0.9155	1000	0.1763	0.9307	1000
HyenaDNA	Ends + Mean + Max	0.0291	0.9155	1000	0.1780	0.9309	1000
ESM-2 650M	—	0.1760	0.9293	1000	0.3503	0.9464	1000
TSS-centred window (196,608 bp; primary = genomic-interval-disjoint split)							
TSS 4-mer	—	-0.0343	0.9091	1000	0.0856	0.9218	1000
DNABERT-2	Mean	-0.0114	0.9111	1000	0.1288	0.9261	1000
DNABERT-2	Max	-0.0413	0.9083	1000	0.0875	0.9221	1000
DNABERT-2	Mean-CLS	-0.0236	0.9099	10000	0.0956	0.9230	1000
DNABERT-2	Ends + Mean	-0.0253	0.9098	10000	0.0934	0.9227	10000
DNABERT-2	Ends + Mean + Max	-0.0203	0.9103	10000	0.1067	0.9240	10000
NT-v2	Mean	-0.0131	0.9109	1000	0.1238	0.9257	100
NT-v2	Max	-0.0351	0.9089	1000	0.0837	0.9217	1000
NT-v2	Mean-CLS	-0.0347	0.9089	10000	0.0466	0.9181	1000
NT-v2	Ends + Mean	-0.0265	0.9097	10000	0.0771	0.9214	1000
NT-v2	Ends + Mean + Max	-0.0236	0.9100	10000	0.0847	0.9217	10000
GENA-LM	Mean	-0.0171	0.9106	1000	0.1125	0.9245	1000
GENA-LM	Max	-0.0229	0.9100	10000	0.0888	0.9221	10000
GENA-LM	Mean-CLS	-0.0201	0.9103	1000	0.1069	0.9239	1000
GENA-LM	Ends + Mean	-0.0313	0.9093	10000	0.0809	0.9215	10000
GENA-LM	Ends + Mean + Max	-0.0309	0.9094	10000	0.0847	0.9219	10000
HyenaDNA	Mean	-0.0373	0.9087	100	0.0905	0.9224	10
HyenaDNA	Max	-0.0330	0.9091	10000	0.0700	0.9203	1000
HyenaDNA	Ends + Mean	-0.0352	0.9089	10000	0.0591	0.9193	1000
HyenaDNA	Ends + Mean + Max	-0.0334	0.9091	10000	0.0567	0.9191	1000
Enformer (whole window)	—	-0.0060	0.9118	100000	0.1340	0.9267	10000

C Confidence Intervals and Robustness Checks

C.1 Seed sensitivity and headline intervals

Table C1: Headline cells across four homology cluster-to-split seeds (42, 1, 7, 123), min–max. Each seed re-selects encoder, pooling and amino-acid k on validation; the TSS row keeps DNABERT-2 and uses the seed's disjoint split. Seeds re-assign the same clusters largest-first, so two seed splits share 34–38% of CDS and 39–50% of TSS test genes (independent splits would share about 15%), so the ranges understate split-to-split variation.

Cell	Macro-F1 (4 splits)	GenePT R^2 (4 splits)
ESM-2 650M	0.926–0.955	0.176–0.191
Best DNA encoder	$0.619^{\dagger}$–0.696	0.057–0.069
AA composition	0.696–0.772	0.072–0.088
TSS (DNABERT-2)	0.217–$0.276^{\dagger}$	-0.013–-0.010

C.2 Additional robustness analyses

Table C2: TSS-window family macro-F1 on the homology-aware split and on a genomic-interval-disjoint split, in which no window overlaps a window in another partition. On the homology split, 212 of the 459 scored test genes (46.2%) share window sequence with a training or validation gene. Encoders at their validation-selected whole-window pooling rule.

Source	Homology F1	Disjoint F1	Δ
Shuffled labels (median of 200)	—	$0.167^{\dagger}$	—
TSS 4-mer	0.283	0.281	-0.002
DNABERT-2	0.361	0.258	-0.103
NT-v2	0.347	0.300	-0.047
GENA-LM	0.306	0.340	+0.033
HyenaDNA	0.258	0.264	+0.006
Enformer (whole window)	0.336	0.413	+0.077
Enformer (central 2,048 bp)	0.402	0.411	+0.009

Table C3: Paired cluster-bootstrap differences on the same test genes (1,000 resamples). The four primary tests carry Holm-adjusted one-sided p; every other block is exploratory with unadjusted p. Blocks: GenePT comparisons on CDS; CDS against the TSS window per encoder (disjoint split); Ends + Mean against Mean repeated three times at three times its C, the matched-regularisation control; the selected encoder against composition plus log CDS length; and the primary tests with each label-noise set masked from scoring (gene counts are the test genes removed).

Comparison	Δ [95% CI]	p
Primary, macro-F1 (Holm-adjusted p)		
NT-v2 (Ends + Mean + Max) − CDS 6-mer	+0.054 [-0.011, +0.126]	0.108
NT-v2 (Ends + Mean + Max) − AA 3-mer	-0.075 [-0.137, -0.001]	0.980
ESM-2 650M − NT-v2 (Ends + Mean + Max)	+0.261 [+0.200, +0.322]	0.004
NT-v2 (Mean (Boundary-including)) (CDS) − NT-v2 (Mean) (TSS) (disjoint split)	+0.393 [+0.312, +0.479]	0.004
Exploratory, GenePT R^2 (unadjusted p)		
AA 3-mer − AA 2-mer	+0.021 [+0.013, +0.029]	0.001
AA 3-mer − CDS 6-mer	+0.042 [+0.031, +0.054]	0.001
DNABERT-2 (Mean (Boundary-including)) − CDS 6-mer	+0.026 [+0.018, +0.034]	0.001
DNABERT-2 (Mean (Boundary-including)) − AA 3-mer	-0.015 [-0.031, +0.000]	0.971
ESM-2 650M − AA 3-mer	+0.096 [+0.082, +0.111]	0.001
ESM-2 650M − DNABERT-2 (Mean (Boundary-including))	+0.111 [+0.087, +0.134]	0.001
CDS − TSS window, same genes, disjoint split, macro-F1 (unadjusted p)		
DNABERT-2	+0.380 [+0.295, +0.452]	0.001
NT-v2	+0.393 [+0.312, +0.479]	0.001
GENA-LM	$+0.267^{\dagger}$ [+0.173, +0.371]	0.001
HyenaDNA	+0.388 [+0.307, +0.455]	0.001
4-mer (control)	+0.268 [+0.195, +0.346]	0.001
CDS − TSS window, same genes, disjoint split, GenePT R^2 (unadjusted p)		
DNABERT-2	+0.060 [+0.046, +0.071]	0.001
NT-v2	+0.038 [+0.027, +0.049]	0.001
GENA-LM	+0.047 [+0.035, +0.056]	0.001
HyenaDNA	+0.052 [+0.040, +0.063]	0.001
4-mer (control)	+0.048 [+0.039, +0.058]	0.001
Ends + Mean − Mean ×3 at $3 \times C$, macro-F1 (unadjusted p)		
DNABERT-2	-0.021 [-0.062, +0.020]	0.838
NT-v2	-0.001 [-0.026, +0.028]	0.509
GENA-LM	$+0.086^{\dagger}$ [+0.017, +0.153]	0.006
HyenaDNA	-0.016 [-0.045, +0.014]	0.862
Encoder − composition with log CDS length, macro-F1		
NT-v2 (Ends + Mean + Max) − CDS 6-mer + length	+0.051 [-0.017, +0.122]	0.064
NT-v2 (Ends + Mean + Max) − AA 3-mer + length	-0.089 [-0.153, -0.020]	0.994
Without noisy TF labels (84 test genes, 79 disjoint), macro-F1		
NT-v2 (Ends + Mean + Max) − CDS 6-mer	+0.047 [-0.025, +0.119]	0.100
NT-v2 (Ends + Mean + Max) − AA 3-mer	-0.076 [-0.140, -0.006]	0.984
ESM-2 650M − NT-v2 (Ends + Mean + Max)	+0.255 [+0.192, +0.323]	0.001
NT-v2 (Mean (Boundary-including)) (CDS) − NT-v2 (Mean) (TSS) (disjoint split)	+0.420 [+0.331, +0.503]	0.001
Without non-protein-kinase labels (23 test genes, 24 disjoint), macro-F1		
NT-v2 (Ends + Mean + Max) − CDS 6-mer	+0.054 [-0.020, +0.133]	0.082
NT-v2 (Ends + Mean + Max) − AA 3-mer	-0.066 [-0.133, +0.001]	0.972
ESM-2 650M − NT-v2 (Ends + Mean + Max)	+0.254 [+0.188, +0.316]	0.001
NT-v2 (Mean (Boundary-including)) (CDS) − NT-v2 (Mean) (TSS) (disjoint split)	+0.392 [+0.314, +0.473]	0.001
Without templated summaries (55 test genes), GenePT R^2		
DNABERT-2 (Mean (Boundary-including)) − CDS 6-mer	+0.024 [+0.018, +0.031]	0.001
DNABERT-2 (Mean (Boundary-including)) − AA 3-mer	-0.022 [-0.032, -0.012]	1.000
ESM-2 650M − DNABERT-2 (Mean (Boundary-including))	+0.119 [+0.104, +0.135]	0.001

Table C4: Ridge-to-GenePT alignment under measures that do not depend on the GenePT basis. Pooled R^2 weights each dimension by its variance; retrieval@5 is the share of the 459 scored CDS test genes (465 for the TSS row, on the disjoint split) whose true GenePT vector ranks in the top five by cosine to the prediction (chance 1.1%); median rank of the true vector.

Method	Macro-R^2	Pooled-R^2	Retr.@5	Med. rank
ESM-2 650M (upper bound)	0.176	0.187	27.5%	18
ESM-2 150M	0.158	0.169	20.3%	31
DNABERT-2, Mean (Boundary-including) (best DNA)	0.065	0.072	5.0%	96
NT-v2, Mean (Boundary-including)	0.043	0.051	6.3%	79
GENA-LM, Ends + Mean	0.049	0.056	5.9%	91
HyenaDNA, Ends + Mean	0.029	0.035	5.2%	111
TSS DNABERT-2, Mean	-0.011	-0.010	2.4%	195
Shuffled-label control (4-mer, median of 200)	-0.027	-0.026	1.1%	230

Table C5: TSS-window family macro-F1 under whole-window pooling (validation-selected rule) versus TSS-Anchored pooling, each with its 95% cluster-bootstrap CI; Enformer as whole-window mean versus central 2,048 bp readout. Δ: anchored minus whole window on the same test genes, paired bootstrap; an anchored cell is bold where that CI lies above 0. Chunk comp.: the validation-selected composition baseline (4-mer + GC or 6-mer) of the same chunk. Shuffled-label median: $0.167^{\dagger}$.

Model	Whole window	TSS-Anchored [95% CI]	Δ [95% CI]	Chunk comp.
Homology-aware split				
TSS 4-mer	0.283 [0.228, 0.331]	—	—	—
DNABERT-2	0.361 [0.276, 0.427]	0.261 [0.214, 0.299]	-0.100 [-0.178, -0.010]	0.293
NT-v2	0.347 [0.284, 0.394]	0.370 [0.311, 0.417]	+0.023 [-0.038, +0.095]	0.304
GENA-LM	0.306 [0.241, 0.365]	0.350 [0.285, 0.408]	+0.044 [-0.021, +0.112]	0.291
HyenaDNA	0.258 [0.215, 0.296]	0.298 [0.231, 0.340]	+0.040 [-0.032, +0.097]	0.329
Enformer	0.336 [0.251, 0.393]	0.402 [0.312, 0.460]	+0.067 [-0.021, +0.150]	—
Genomic-interval-disjoint split				
TSS 4-mer	0.281 [0.223, 0.331]	—	—	—
DNABERT-2	0.258 [0.199, 0.308]	0.281 [0.235, 0.322]	+0.024 [-0.030, +0.075]	0.301
NT-v2	0.300 [0.239, 0.354]	0.351 [0.285, 0.411]	+0.050 [-0.005, +0.106]	0.336
GENA-LM	0.340 [0.266, 0.406]	0.367 [0.304, 0.427]	+0.027 [-0.058, +0.120]	0.262
HyenaDNA	0.264 [0.220, 0.305]	0.276 [0.224, 0.330]	+0.012 [-0.055, +0.079]	0.325
Enformer	0.413 [0.331, 0.487]	0.411 [0.345, 0.472]	-0.001 [-0.047, +0.044]	—

Table C6: Partition composition: genes, the share in singleton 40% protein clusters, the median cluster size, the share in clusters of 10 or more, and olfactory receptors among GPCRs. A cluster split with family quotas places the largest clusters in training, so test is mostly genes without close paralogs. Part of the random-split gain in §3.5 may therefore come from this different test population, not only from leaked homologs.

Partition	Genes	Singleton	Median	In $\geq$10	ORs / GPCRs
Homology-aware split (CDS primary)					
Train	2,271	19%	3	33%	388 / 414
Val	486	87%	1	0%	7 / 88
Test	487	87%	1	0%	6 / 89
Genomic-interval-disjoint split (TSS primary)					
Train	2,271	29%	3	31%	346 / 414
Val	487	61%	1	11%	55 / 88
Test	486	62%	1	0%	0 / 89

C.3 TSS-window genomic-overlap audit

To confirm that the TSS windows probe a substrate distinct from CDS, we measured how much of each 196,608 bp window overlaps the target gene's own coding sequence, using the GRCh38 Ensembl release 115 gene annotation. Across all 3,244 windows the mean target-CDS fraction is 0.008 (∼1%). The rest is the gene's own introns (18%) and untranslated exons (1%), the exons (11%) and introns (44%) of neighbouring genes, and intergenic sequence (25%). Because so little of the gene's own coding sequence leaks back into the window, the substrate ablation is interpreted on the unmasked window, with no explicit coding mask applied.

C.4 Split leakage audit

To measure what a random split leaks, we counted the test genes whose 40% protein cluster also contains a train or validation gene. On the random split 271 of 487 test genes (56%) have such a relative; on the homology split, none do by construction. The score gap between the two splits is reported in §3.5 and Table B4.

D Code Reproducibility

The code is at `github.com/Austin-Senna/dna_to_text` under the MIT licence; data and results are CC BY 4.0, except files derived from NT-v2 outputs, which keep NT-v2's CC BY-NC-SA 4.0. The tag `mlcb2026-camera-ready` marks the code behind every number in this paper, with every dependency pinned. Embedding extraction needs a GPU; every later step runs on CPU. A single script refits every probe cell from a clean tree, and an independent minimal reimplementation (NumPy, pandas, scikit-learn) reproduces the headline cells from the stored features. Sequences, embeddings, chunk caches, stored test predictions, the gene table and the HGNC snapshot it was built from are deposited on Zenodo (10.5281/zenodo.23031890), so every step after extraction can be rerun from the deposit without a GPU.

D.1 Deterministic settings and hyperparameters

Splits. The primary homology-aware split: each gene's canonical CDS is translated and clustered with MMseqs2 `easy-cluster` at 40% identity and 80% bidirectional coverage (`-min-seq-id 0.4 -c 0.8`), and whole clusters are assigned to partitions by a greedy largest-cluster-first rule that keeps 70/15/15 family proportions (seed 42). The split holds 2,271 train, 486 validation and 487 test genes (1,751 clusters over 3,244 genes); seeds 1, 7 and 123 give the replicate splits. The TSS-disjoint split groups genes that share a protein cluster or an overlapping window. Before scoring, a validation or test gene is masked when an all-against-all MMseqs2 search on full-length proteins finds a partner in an earlier partition at $E \leq 10^{-3}$, ≥40% identity and ≥80% coverage of both sequences (on the disjoint split, also an overlapping window); training genes are never masked.

Probes. Features are standardised inside every fit, with the scaler fit on the rows being fit. Classification is scikit-learn logistic regression (`lbfgs`, `tol=1e-6`, `max_iter=5000`, no class weighting); regression onto GenePT is ridge regression on unscaled targets. C and α start on the grid $\{10^{-4}, \ldots, 10^{4}\}$, which extends one decade at a time while the validation pick sits on its edge, up to 10^{-6} and 10^{6} for C and 10^{9} for α. Extension stops when a decade gains no more than 10^{-4} or a fit fails to converge. A few cells, none behind a primary test, selected a value adjacent to a non-converged fit and keep their validation pick. Each hyperparameter, pooling rule, composition k (nucleotide 4 or 6; amino acid 1, 2 or 3) and encoder is chosen on validation (macro-F1, or macro-R^2 for GenePT), separately on every split; the probe is then refit on train+validation and scored once on test. Every fit runs on one BLAS thread. A refit of the whole protocol on 6 threads, and another on a different OpenBLAS kernel, changes 44 of 764 cells (19 select a different C; test macro-F1 moves by at most 0.042) and no primary test; those digits carry a dagger in the appendix.

Statistics. Statistics are computed from the stored test predictions without refitting. Intervals and one-sided paired p-values come from 1,000 bootstrap resamples (seed 42) of whole groups: 40% protein clusters for CDS cells on the homology split, window-and-protein groups for TSS cells and on the disjoint split, joined through every protein pair that passes the purge rule and, for GenePT, through genes that share a templated summary. Resampling is by group, not stratified by family. Holm correction covers the four primary tests; every other test is exploratory and unadjusted. Each null band is the central 95% of 200 refits with train and validation labels shuffled (seeds 0–199), each re-selecting the regularisation strength; test labels are never shuffled.

Translation. Amino-acid k-mers and ESM-2 read each CDS translated through internal stops (as X), dropping a terminal stop and any trailing 1–2 nucleotides; the clustering behind the splits translates to the first stop. In translation a codon with a non-ACGT base becomes X, and amino-acid k-mers containing X are skipped; the codon baseline skips such codons. Of the canonical CDS, 18 are irregular: 17 carry an internal stop (14 of them olfactory receptors) and 1 an ambiguous base.

Extraction. Models run frozen, in inference mode, at revisions pinned in the code. CDS chunks are 510 content tokens overlapping by 64 for DNABERT-2 and GENA-LM, 998 overlapping by 64 for NT-v2, and 8,192 overlapping by 512 for HyenaDNA, which has no boundary tokens. TSS windows are 196,608 bp in gene orientation, centred on the canonical transcript's 5′ end (Ensembl release 115); windows that run past a chromosome end are padded with N (8 genes, 498 to 54,023 bp), the same input for every model. Enformer's whole-window summary averages all 896 output bins of 128 bp (the central 114,688 bp; the outer flanks are attention context only), and its central summary averages the central 16 bins (2,048 bp).

E Changes Since Submission

The camera-ready reruns every probe under a corrected protocol. We fixed the four primary tests (§3) after we had seen, on the same test genes, the submitted results and early reruns of a few cells under corrected settings, and before the first full run of the corrected protocol. After that trial run, an audit changed how the bootstrap groups related genes (Appendix D.1); this moved the tests' intervals and Holm p-values but not their point estimates, and changed no test's outcome. The tests are therefore not pre-registered. The submitted version had the following problems; correcting them changed the numbers (Table E1):

- **Pooling was selected on test,** not on validation as the submitted Methods said (the headline cell, NT-v2 with Ends + Mean + Max, was such a pick), and DNABERT-2 was fixed as the TSS representative by an earlier test pick. Every choice is now made on validation.
- **GENA-LM ran untrained.** Under the library version we used, GENA-LM's pretrained weights silently failed to load, so every submitted GENA-LM number came from a randomly initialised network. The pipeline now checks the loaded parameters against the weights file.
- **HyenaDNA read an untrained token.** A CLS token HyenaDNA was never trained with sat at the start of every input, so every HyenaDNA rule read it, and its CLS feature was the same for every gene. HyenaDNA now reads raw sequence, and boundary-token rules are undefined for it.
- **TSS windows.** Windows of minus-strand genes were not reverse-complemented, and the outermost gene start stood in for the TSS. Windows are now in gene orientation, centred on the canonical transcript's 5′ end.
- **Enformer.** Tables mixed its readouts; it is now reported as the mean of all 896 output bins (whole window) and of the central 16 (central readout).
- **Translation.** The 17 CDS with an internal stop were cut at it for the amino-acid baselines

and ESM-2; they now translate through it.

- **Probes.** Every probe was fit on unscaled features, logistic fits stopped at tolerance 10^{-4} or 2,000 iterations, results depended on the BLAS thread count, composition baselines were picked at the edge of the regularisation grid, and the nucleotide k was fixed at 4. Features are now standardised inside every fit, grids extend past their edge, both k are chosen on validation, and logistic fits run on one thread to tolerance 10^{-6} and up to 5,000 iterations (settings in Appendix D.1).
- **Evaluation.** The submission resampled genes individually, applied no multiple-comparison correction, and estimated chance from one label shuffle. Intervals now come from a cluster bootstrap with Holm adjustment over the four primary comparisons, chance is a band from 200 shuffles, test genes with a close homolog in an earlier partition are masked from scoring, and the TSS arm uses a split that also separates overlapping windows.

These submitted conclusions changed:

- the selected encoder's lead over nucleotide composition, which the submission called well clear, is not significant (Holm $p = 0.108$), and it does not reach amino-acid composition, which the submission called level with it;
- GENA-LM, which the submission placed below 4-mer composition, scores highest of the four encoders on test once its weights load (0.710 on CDS, 0.340 on the TSS window), though it is not the validation pick;
- the submission read the ranking of the rules that add the first and last chunks as positional structure; at matched regularisation they show an exploratory gain only for GENA-LM (Section 3.3);
- on GenePT, amino-acid 3-mer composition, above every DNA encoder in the submission, is no longer reliably above the selected encoder: the paired interval touches zero (Appendix Table C3);
- on the TSS window, the encoders no longer clear composition: the submitted table put all four above the 4-mer (0.244 to 0.326 vs. 0.181); they now sit around it (0.258 to 0.340 vs. 0.281);
- the TSS window sits above chance. The submission's single-shuffle reference (0.224) was high; the 200-shuffle median is $0.167^{\dagger}$. The encoders' margin over chance is about what it was, now on the disjoint split (homology-split values in Table C2), and the window stays far below CDS on the same genes (§3.4.1).

Table E1: Submitted versus camera-ready values for the main quantities: the submitted value, this version's value, and the cause; full values are in Appendix B. CDS rows are on the homology-aware split except the random-split rows; TSS rows were submitted on the homology-aware split and are now on the disjoint split. Macro-F1 unless marked R^2. "Selected" is the submission's best-on-test cell and this version's validation pick. Two causes apply to every row and are not repeated: the corrected probe protocol (standardised features, extended grids, tighter convergence, one thread) and, except on the random split, masking test genes with a close homolog. Row causes: P, pooling and encoder selected on validation; K, composition k chosen on validation; T, translation through internal stops; G, GENA-LM weights loaded; H, HyenaDNA's untrained CLS token removed; W, TSS windows re-oriented and re-centred; E, Enformer readouts separated; S, cluster bootstrap with Holm adjustment; C, chance from 200 shuffles; D, disjoint split; —, the two shared causes only.

Quantity	Submitted	This version	Cause
Family classification, CDS			
Selected DNA encoder (NT-v2)	0.727	0.693	—
NT-v2 − nucleotide comp.	well clear	+0.054 [−0.011, +0.126]	K, S
NT-v2 − amino-acid comp.	level	−0.075 [−0.137, −0.001]	K, T, S
Nucleotide composition reference	0.633 (4-mer)	0.639 (6-mer)	K
Amino-acid composition reference	0.735 (AA 2-mer)	0.767 (AA 3-mer)	K, T
ESM-2 650M	0.960	0.954	T
DNABERT-2	0.654	0.628	P
HyenaDNA	0.641	0.633	H, P
GENA-LM (submitted: untrained)	0.415	0.710	G, P
NT-v2, pooling range	0.519–0.727	0.504–0.698	—
Chance	0.224 (one shuffle)	0.167 (median of 200)	C
GenePT regression, CDS (R^2)			
Selected DNA encoder (DNABERT-2)	0.077	0.065	P
AA 3-mer	0.090	0.080	T
CDS 4-mer	0.045	0.032	—
ESM-2 650M	0.181	0.176	T
Family classification, TSS window			
Selected TSS encoder (DNABERT-2)	0.326	0.258	W, D, P
NT-v2	0.313	0.300	W, D, P
GENA-LM (submitted: untrained)	0.244	0.340	G, W, D, P
TSS 4-mer	0.181	0.281	W, D
Enformer central readout	0.401	0.411	W, E, D
Random split			
NT-v2, CDS	0.828	0.801	P
AA 2-mer, CDS	0.837	0.854	T
NT-v2, TSS	0.447	0.487	W, P